

Variable scope

Different types of [variables](#) are

- Local Variables
- Instance Variables
- Static Variables

Local Variables A variable defined within a block or method or constructor is called a local variable. • These variables are created when the block is entered, or the function is called and destroyed after exiting from the block or when the call returns from the function. • The scope of these variables exists only within the block in which the variables are declared, i.e., we can access these variables only within that block. • Initialization of the local variable is mandatory before using it in the defined scope.	<pre> Class hcl guvi { public static void main(String[] args) { // Declared a Local Variable int var = 10; // This variable is local to this main method only System.out.println("Local Variable: " + var); } } </pre>
Instance variables are non-static variables and are declared in a class outside of any method, constructor, or block. • As instance variables are declared in a class, these variables are created when an object of the class is created and destroyed when the object is destroyed. • Unlike local variables, we may use access specifiers for instance variables. If we do not specify any access specifier, then the default access specifier will be used. • Initialization of an instance variable is not mandatory. Its default value is dependent on the data type of variable. • Instance variables can be accessed only by creating objects. • We initialize instance variables using constructors while creating an object. We can also use instance blocks to initialize the instance variables.	<pre> // Java Program to demonstrate class hcl guvi { // Declared instance variable String student = "Shubham Jain"; public static void main(String[] args) { HCL Guvi Zen=new HCLGuvi(); // geek variable can be accessed by creating object System.out.print(Zen. student); } } </pre>

Static Variables

Static variables are also known as class variables.

- These variables are declared similarly to instance variables. The difference is that static variables are declared using the static keyword within a class outside of any method, constructor, or block.
- Unlike instance variables, we can only have one copy of a static variable per class, irrespective of how many objects we create.
- Static variables are created at the start of program execution and destroyed automatically when execution ends.
- Initialization of a static variable is not mandatory. Its default value is dependent on the data type of variable

```
// Java Program to demonstrate class
hcl guvi {
// Declared instance variable Public
static String student = "Shubham Jain";
public static void main(String[] args)
{
// student variable can be accessed without
object
// static variable
System.out.print(hclguvi. student);
}
}
```

Scope of variables

Variable Type	Scope	Lifetime
Instance variable	Troughout the class except in static methods	Until the object is available in the memory
Class variable	Troughout the class	Until the end of the program
Local variable	Within the block in which it is declared	Until the control leaves the block in which it is declared